

MATH 3430 - Week 1

First-Order Equations

1 Overview

We start by studying **first-order differential equations**. Certain special types will be considered (**separable, autonomous, linear**).

We will approach these equations (and the other equations we meet later in the course) three different ways:

1. Analytically - developing techniques to find solutions, either given as explicit or implicit functions.
2. Qualitatively - developing techniques to understand the geometric character of solutions, without necessarily finding analytical solutions.
3. Numerically - using Sage to find approximate solutions and plotting them.

An important aspect of the course will be seeing how differential equations can **model** many important “real-world” phenomena. We will talk about how these models are built and how to interpret the solutions that we get (by any of the three methods given above).

2 Schedule and Checklist

Here is a brief schedule for the week. Please note that this is subject to change, so you need to adjust to what we do in class, regardless of what it says here. Think of this as a pretty accurate guideline, though.

- (M) **No Class!**
- (T) Introduction to the course. Modeling. Unlimited Population Growth. Constrained Population Growth.
- (W) Separable Equations. Linear Equations and Integrating Factors.
- (R) Slope Lines. Equilibrium Solutions. Stability. Phase Lines.
- (F) **No Class!**

Here is a checklist of things to do this week. This is meant to help you keep track of what you’ve completed and what you still need to work on.

- ☐ Go to the Google Drive folder for the class

<https://drive.google.com/drive/folders/1PvHpPe9SAjudNsYy9L1EeVhCVHq3IRW7>

and look around.

- ☐ Look through these notes and ask any questions that you have about anything.
- ☐ Read the sections from Judson and Lebl.
- ☐ Complete all of the homework exercises.

3 Terminology and Notation

- First-Order Differential Equations
 - Separable Equations
 - Linear Equations
 - Nonlinear Equations
 - Autonomous Equations
 - Homogeneous and Nonhomogeneous Equations
- General Solution
- Initial Conditions and Initial Value Problems (IVP)
- Existence and Uniqueness Theorems
- Integrating Factors
- Direction Fields
- Equilibrium Solutions
- Phase Lines
 - Sink
 - Source
 - Node (Saddle)

4 Equations and Models

- Unlimited Population Growth (Exponential Population Model). Let k be the growth-rate parameter.

$$\frac{dP}{dt} = kP$$

- Constrained Population Growth (Logistic Population Model). Let k be the growth-rate parameter and N the carrying capacity.

$$\frac{dP}{dt} = kP \left(1 - \frac{P}{N} \right)$$

- Separable equation.

$$\frac{dy}{dt} = g(t)h(y)$$

- First-Order Linear Nonhomogeneous Equations.

$$y' + p(t)y = g(t)$$

- First-Order Autonomous Equations.

$$\frac{dy}{dt} = f(y)$$

5 Important Formulas (analytical material)

- The integrating factor for a linear equation $y' + p(t)y = g(t)$ is given by:

$$\mu(t) = \exp\left(\int p(t)dt\right)$$

- (Existence and Uniqueness Theorem for First-Order Linear Diff. Eqns.) If

$$y' + p(t)y = g(t), \quad y(t_0) = y_0$$

is an IVP and $p(t)$ and $g(t)$ are continuous on the interval $I = (\alpha, \beta)$ (α can equal $-\infty$ and/or β can equal ∞), then there exists a unique function $y = \phi(t)$ that solves the IVP on I .

- (Existence and Uniqueness Theorem for First-Order Differential Equations) If $f(x, y)$ is continuous and $\partial f / \partial x$ exists and is continuous near (x_0, y_0) , then a solution to

$$y' = f(x, y), \quad y(x_0) = y_0$$

exists (at least for x in some small interval centered around x_0) and is unique.

6 Geometric and Visual Aspects (qualitative material)

- If a differential equation satisfies the conditions of the Existence and Uniqueness Theorems, then solution curves **can not** intersect (if they did, this would violate the uniqueness).
- The direction field for an autonomous equation is completely determined by a single vertical line (the phase line). Once the behavior of this line is understood, the rest of the direction field is just horizontal translates of it.
- Equilibrium solutions for autonomous equations will always be horizontal lines. By the uniqueness theorem, any solution curve that approaches equilibrium will never actually reach it but will always be asymptotic to it. This will look like a horizontal asymptote.
- Solutions can **blow up** to infinity in finite time. An example of this is given by $y' = 1 + y^2$, $y(0) = 0$ (the solution is $y(t) = \tan(t)$, which has vertical asymptotes at $t = \pm \frac{\pi}{2}$. Please note that this is a nonlinear equation, so the stronger Existence and Uniqueness Theorem for Linear Equations does not hold here.
- Here is an example of a direction field for the Logistic Growth Model with Harvesting given by the differential equation

$$P' = \left(1 - \frac{P}{200}\right) - 32$$

Notice the asymptotic approach to the equilibrium solution at $P(t) = 160$ (there is a second equilibrium solution at $P(t) = 40$ but the solutions curves move away from it).

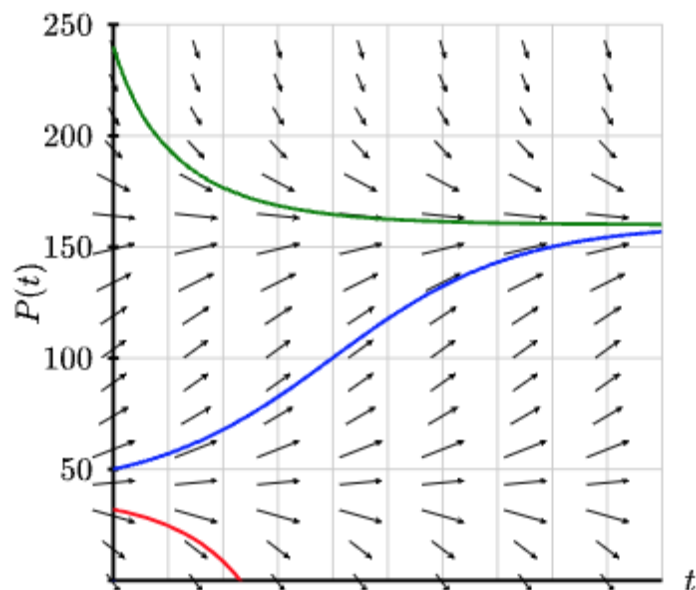


Figure 1: Direction Field for Logistic Growth with Harvesting

- Here is an example of a phase line for the same Logistic Growth Model with Harvesting from the previous example. Note that the equilibrium solution at $P = 160$ is a **sink**, so solution curves move towards it, while the equilibrium solution at $P = 40$ is a **source**, so the solution curves move away from it.

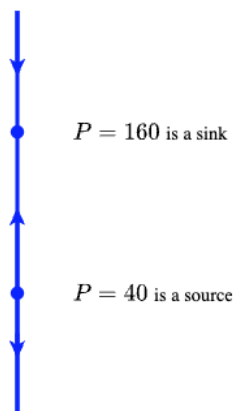


Figure 2: Phase Line for Logistic Growth with Harvesting

7 Sage Code (numerical material)

7.1 Commands

There are a lot of commands here. Later weeks won't have as many. Think of this more as a reference. The best way to learn how to actually use the commands is to look at the commands in actual code. There are some examples below.

- `var('variable name')`

Declares the given variable(s).

- `function('function name')(variable)`

Defines the named function to be a function of the given variable(s).

- `diff(function, variable, n)`

Differentiates the given function with respect to the variable. If there is a value for n , the function is differentiated n times. If there is no value for n , the function is differentiated once.

- `desolve(diff eqn, [function, variable], ics=[ind variable initial value, dep variable initial value])`

Solves the given initial value problem with respect to the variable. If the `ics=[t_0, y_0]` command is not included, gives the general solution to the differential equation.

- `.show(options)`

Use this command after a computed object or graph to display it.

- `plot(function, (variable, min value, max value))`

Creates a graphic object that is the graph of the given function with respect to the variable.

- `plot_slope_field(function, (horizontal variable, min, max), (vertical variable, min max))`

Plots the slope field (direction field) for the given function with respect to the horizontal variable (usually t) and the vertical variable (usually x or y). All of the standard plot options (like *thickness*, *color*, *labels*, etc.) can be used with this.

- `desolve_rk4(diff eqn, dep variable, ivar = ind variable, ics=[ind variable initial value, dep variable initial value], end_points=[ind variable min, ind variable max], step=step size)`

Numerically solves the given IVP, where *diff eqn* specifies the differential equation, *ics* specifies the initial condition, for values of the independent variable (specified by *ivar*) in the range specified by *end_points*. The independent variable changes by steps of size *step*. In general, the smaller this number is, the more accurate the approximation is, but at the cost of needing more computations and so longer run time. Usually a step size of around 0.1 is adequate, though sometimes you will want this to be smaller. The output of this command is a list of points.

- `point((x0, y0))`

Plots a point with coordinates (x_0, y_0) .

- `line([(x1, y1), (x2, y2)])`

Plots a line from the point (x_1, y_1) to (x_2, y_2) .

7.2 Example Code

- Solving $y'(t) = y - t$:

```
t = var('t')
y = function('y')(t)
de = diff(y,t)== y - t
solution = desolve(de,[y,t])
solution.show()
```

- Solving the IVP $y'(t) = y - t$, $y(0) = 0$ and plotting the solution curve:

```
t = var('t')
y = function('y')(t)
de = diff(y,t)== y - t
solution = desolve(de,[y,t], ics=[0,0])
solution.show()
p = plot(solution, (t,-3,3))
p.show(ymin=-3.5, ymax=3.5)
```

- Plotting the direction field for $y'(t) = y - t$ for $-3.5 \leq t \leq 3.5$ and $-3.5 \leq y \leq 3.5$:

```
t, y = var('t y')
f(t,y) = y - t
p = plot_slope_field(f, (t, -3.5, 3.5), (y, -3.5, 3.5))
p.show()
```

- Numerically solving the IVP, $y'(t) = y - t$, $y(0) = 0$, and plotting the solution curve over the direction field:

```
t, x = var('t x')
y = function('y')(t)
f(t, x) = x - t
de = diff(y,t)== y - t
solution = desolve(de,[y,t], ics=[0,0])
solution.show()
p = plot(solution, (t,-3,3))
p += plot_slope_field(f, (t, -3, 3), (x, -3.5, 3.5))
p.show(ymin=-3.5, ymax=3.5)
```

8 General Comments

- There's a lot going on here. That's a blessing and a curse. It's a blessing because it means that the techniques and the concepts behind them are very powerful. It's a curse because there's so much going on that sometimes things can get lost in the mix. Put the time to understand what's happening, both on the surface and as far below it as you can get. At first it won't be very far, but stick with it. It's worth it.
- Don't be surprised or dismayed if something that you think you understood yesterday all of sudden becomes lost again. That's how these things work. But each time you re-understand something, you'll understand it a little bit better. It will become more fully realized, more fleshed-out. That's the end goal. It won't happen right away. So don't expect it to. You get better at something not by doing it

right once, but by doing it over and over again. Steph Curry didn't stop practicing his threes after he became the best shooter in NBA history. If anything, he started practicing more. And here's something else to keep in mind: Steph Curry has missed more three pointers than anyone in NBA history (other than James Harden). It's like the Great One said: 'You miss 100% of the shots you don't take.' Don't be afraid to get something wrong. Take the shot, give it a try. You'll get better.

- Being able to solve a differential equation is only one of three main steps (if not more). The first is understanding what the differential equation is saying by itself, what the variables represent, what the constants/parameters mean, why the derivatives that show up are showing up, what the initial conditions are and what they mean. All of that happens before you even attempt to solve the equation. Then you have to 'solve' the equation. Even this can mean a couple of different things. Are you able to get an explicit solution? An implicit solution? A numerical solution? A qualitative solution? All of these are important and can be enough. Not all of them will always be possible. Once you have your 'solution,' then you still have to try to understand what it is telling you (and what it's not). If your answer is a function, what does the function tell you about the situation being modeled by the differential equation? If your answer is a graph, what information can you 'read' from the graph? If your answer is numerical, what do those numbers mean and what are you going to do with them? All of this happens after you've solved the equation. All of these things need to be taken into consideration. It's about so much more than just writing down an answer. I'd much rather you be able to intelligently address all of these issues and struggle with getting the 'right' answer than be able to write down the answer but not take any time to think about where the equation came from, how you solved it, and what it all means at the end of the day.
- Use any and all tools that you can to make progress with these questions. Take the time to learn Sage, at least the parts of it that arise in this class. You will not regret it. In the 'real world,' no one actually solves differential equations by hand. There's just too much going on for that to be practical. But you do need to have a firm grasp of the underlying concepts and techniques to be able to fully understand what's going on, especially in regards to the first (formulation of the differential equation) and third (interpretation of the solution) aspects mentioned in the previous paragraph.
- Don't be afraid to ask questions. A good question is usually more important than a good answer.
- Please read all of the notes, and read them again (and again). Write on them. Underline things. Make your own notes on the notes. Draw pictures. Do something. That being said, these notes will not replace, nor are they meant to replace, active participation in class. This means coming to class every day, taking notes on what we go over, asking questions, coming to office hours, etc. Similarly, don't wait to get started on the homework. Get to it right away. If you're not able to do something, the sooner you know that the more time you have to rectify the situation. Again, questions and office hours are key. I will not spend much time in class going over homework problems. There's too much other stuff that we need to do with that time. But I am more than happy to go over anything connected to the homework in office hours.

9 Reading Notes

The main reference are the notes by Judson. Chapters and sections from his ebook will be denoted either by (J $N.M$), for chapter N , section M , or (J $N.M.S$), for chapter N , section M , subsection S . The other reference is Lebl. The same notation will be used but will look like (L $N.M$), and so on. Occasionally, it will say something like "J Example 1.3.2", which means exactly what you think it means (that the appropriate reference is found listed as example 1.3.2 in Judson).

This is **not** comprehensive. It is always a good idea to look things up in the index and to explore.

- Population Growth Models
 - Exponential Growth (J 1.1.1)

- Logistic Growth (J 1.1.2)
- First-Order Equations
 - Separable Equations (J 1.2) (L 1.3)
 - Autonomous Equations (J 1.3.5) (L 1.6)
 - Linear Equations (J 1.5.2) (L 1.4)
- Integrating Factors (J 1.5.2) (L 1.4)
- Existence and Uniqueness Theorems (J 1.5.5, 1.6.1) (L 1.2.2)
- Direction Fields (J 1.3.2, 1.3.4) (L 1.2.1)
- Equilibrium Solutions and Stability (J 1.3.5) (L 1.2.1)
- Phase Lines (J 1.3.5) (L 1.6)

10 Additional Topics

Look in the indices of Judson and Lebl for these topics. Not every topic is covered in both texts, though all of the topics are in at least one of them. Do **NOT** do a general internet search, or even go to something like Wikipedia. All of these topics will return thousands (hundreds of thousands) of pages of results and most of them will be way more advanced than what you will need. Please stick to the texts that we are using for the course.

- Bernoulli Equations
- Exact Equations
- Substitution Methods
- Homogeneous Equations
- Mixing Models
- Newton’s Law of Cooling
- Picard Iteration

11 Homework Exercises

There are two types of homework exercises: 1) **To Turn In** and 2) **For Practice**. As the names imply, the ‘To Turn In’ problems will be turned in, but the ‘For Practice’ problems are just for practice and will not be turned in. All of the ‘For Practice’ problems have answers, but there is no work showing how the answers are arrived at. This will give you a chance to practice and check your answer to make sure you are doing things correctly.

Please do **ALL** of the ‘To Turn In’ exercises. Not all of them will be graded for correctness, but they will all be checked for completion. Failure to do all of them will lower the completion portion of the Homework score.

Please write your solutions up clearly and cleanly, using full English sentences. Try not to have a bunch of arrows pointing from one thing to another or a bunch of things crossed-out and a bunch of other things circled, etc. Presentation matters.

You are encouraged to use Sage as you work through these questions. However, you **MUST** show your work and justify the steps that you take to get to your answer. It is totally acceptable to use Sage to verify your answer, but you must then work through the details yourself and write them up in your solutions. Occasionally, some questions will require you to use Sage. I will be sure to point these out when they arise.

11.1 To Turn In

- Pick one item from the Additional Topics and write a paragraph or two about it. This doesn't have to be extensive, but it should give enough information to explain what the topic is, how it is used, what ideas are involved, etc. As mentioned in the 'Additional Topics' section, please only use Judson and Lebl for this. The terminology and notation should match what is found there, not from Wikipedia or a page from a Google search.
- Judson 1.1) 12, 13, 18, 20, 21

These are pretty routine and shouldn't take very long; that's why there's so many of them.

- Judson 1.2) 13, 18

These are more computational and will require you to integrate some basic functions. These are only 'Calc 1' integrals, so you won't need to use any of the more advanced techniques from Calc 2. These are IVPs, so you will need to solve for C . If you can write your answer explicitly, please do so, but if not, you can leave it as an implicit function (but still be sure to solve for C).

- Judson 1.3) 9, 10, 11, 21, 23

These are geometric in nature and don't involve much computation (or even require you to solve the differential equations).

- Judson 1.5) 14, 30 (only parts a and d)

Number 14 is a straight-up integrating factor question, which will require some computation (integration). Question 30 is a little more theoretical and is probably the hardest question in this homework set. Do yourself a favor and rewrite the right-hand side in terms of $y - 4t$. This will involve factoring a quadratic, but it should be pretty obvious what you have to do. It's also a good idea to move the 4 over to the left-hand side.

11.2 For Practice

- Lebl 1.2) 101, 102, 104
- Lebl 1.3) 101, 102, 103, 107
- Lebl 1.4) 101, 104
- Lebl 1.6) 101, 103, 104